

A Comprehensive Smart Parking System in Context of Bangladesh: Bilingual License Plate Detection and Real-Time Space Availability Using IoT and WSN

Md. Motaharul Islam
Computer Science and Engineering
United International University
Dhaka, Bangladesh
motaharul@cse.uiu.ac.bd

Tinon Turja Majumder
Computer Science and Engineering
United International University
Dhaka, Bangladesh
tmajumder2410048@mscse.uiu.ac.bd

Alamin Refat
Computer Science and Engineering
United International University
Dhaka, Bangladesh
tmajumder2410048@mscse.uiu.ac.bd

Abstract—The following document proposes an IoT-based smart parking system designed to better manage parking by recognizing English and Bangla car license plates. The system utilizes Wireless Sensor Networks (WSN) to detect available parking spaces in real time and offers a mobile application for drivers to easily locate these spots. Convolutional Neural Network (CNN) methods are integrated for number plate recognition, allowing the system to identify both English and Bangla license plates, thereby promoting inclusivity in urban environments. Our solution aims to reduce parking search time and alleviate traffic congestion. The mobile application provides drivers with real-time updates on nearby available parking spaces, facilitating seamless navigation. Experimental results demonstrate the system's capacity for accurate license plate detection and reliable parking space monitoring, contributing to improved urban mobility and parking management.

Keywords—IoT, WSN, CNN Methods, Smart Parking, Dual-language License Plate Recognition, Mobile Application, Urban Traffic Management.

I. INTRODUCTION

Rapid urbanization in cities, particularly in developing countries like Bangladesh, has significantly increased the number of vehicles, exacerbating traffic congestion and parking-related issues. As cities expand, efficient parking management becomes crucial to reduce congestion, save time, and minimize fuel consumption. Traditional parking systems often lead to inefficiencies, with drivers wasting time searching for parking spaces, increasing pollution and road traffic.

Smart parking systems powered by the Internet of Things (IoT) offer a solution by optimizing parking space management. By integrating IoT with Wireless Sensor Networks (WSN), these systems can detect and monitor parking space availability in real time, relaying this information to drivers via mobile applications. One of the critical features of modern parking systems is license plate

recognition, which facilitates parking access and management automation.

This paper presents a smart parking system that leverages IoT, WSN, and Convolutional Neural Network (CNN) methods for license plate recognition. Our system is unique in detecting English and Bangla license plates, ensuring inclusivity for vehicles commonly found in Bangladesh. The mobile application component provides real-time updates on available parking spaces, allowing drivers to find and navigate to open spots quickly. Using CNN enhances the system's accuracy in detecting vehicle plates under various conditions.

Our system addresses critical urban parking challenges by incorporating IoT for real-time parking monitoring, WSN for space detection, and CNN for bilingual license plate recognition. The proposed solution aims to reduce traffic congestion, enhance parking efficiency, and improve the overall urban mobility experience. An automated number plate detection and recognition system without any human interaction has been proposed in the work. The digital number plates have been extracted and detected first with the mathematical morphological-based computer vision approach, and you only look once (YOLOv3) algorithm—The YOLOv3 algorithm performed better than the former approach concerning accuracy and detection frame rate. Tesseract optical character recognition (OCR) and a trained convolutional neural network (CNN) model have been used for character recognition. The creativity of the work belongs to the application and study of multiple approaches in recognizing the number plate, as well as using a diverse dataset containing images of cars from different cities and vehicle classes.

Real-Time Space Allocation:

IoT sensors, such as ultrasonic sensors, are deployed to monitor the availability of parking spots in real time. The IoT sensors continuously track whether the assigned parking spot is occupied.

Duration Tracking:

The IoT sensors continuously track whether the assigned parking spot is occupied.

II. RELATED WORK

In this section, relevant works that have been done regarding smart parking and its various elements have been discussed. In [1], authors use IoT devices associated with cloud computing technologies that offer unlimited storage facilities. It helps to overcome the lower processing abilities of IoT nodes. However, this work involves a Raspberry Pi three-camera to record video to track the detection of vehicles. High-definition video recording takes up tremendous memories and is challenging to process in real-time.

Deep learning or deep structured learning is a new area of Machine learning. Deep learning can identify numbers from an image [2]. They have implemented the Convolutional Neural Network(CNN) algorithm using TensorFlow, which shows 98.5% accuracy in detecting a license plate. Their approach is meticulously done with several considerations, but their model doesn't suit the context of Bangladesh. Several vehicles in Bangladesh have license plates written in Bangla.

In [8], the authors develop a webpage to show the empty spaces of the parking area in a real-time monitoring system. A notification via message will pop up to indicate that the parking area is full. They have used two types of sensors for outdoor and indoor parking areas. However, taking data on the parking spaces is a better approach than other systems, but updating or managing data on a webpage is a painful job, and using two types of sensors will be costly.

A proposed model [6] uses some IR sensors and an embedded controller for real-time data collection. They develop a database server to store and update data on parking spaces. They use a Direct Current (DC) motor at the entry gate to allow access for the Identity Document (ID) holders. Using a DC motor to allow access is a beneficial approach for security purposes, but the model will not perform well when there are too many cars simultaneously.

While only considering the Bangla license plate, the authors in the proposed model [bangla license plate] are not vigilant about the sensor that could be used. Our proposed model has scrupulously overcome all of these issues that have been mentioned.

III. PROPOSED METHODOLOGY

Functions with adequate steps that are necessary for setting up the whole process are mentioned below:

A. The Cloud:

The IBM MQTT server is hosted on the cloud. The cloud functions as a database to store all records pertaining to parking areas and authorized users of the system. It meticulously tracks each user connected to the system and maintains essential information, such as the time of car parking, duration of parking, user payment, and payment method. The adaptable nature of the cloud permits the system to seamlessly add any number of users at any time. Additionally, continuous data backups are performed to ensure swift and efficient recovery in the event of a system failure.

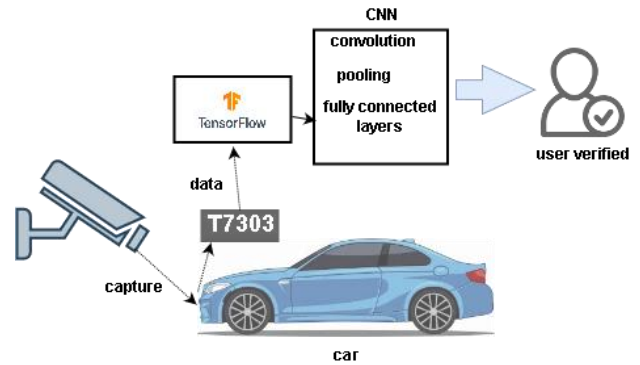


Figure 1: Smart Parking System [2]

B. License Plate Detection

The license plate should be detected first to classify the car. It could be done in the following processes:

Method 1: Computer Vision & Morphological Operation

First, the number plate is separated and cropped from the image. The input from the user could be any type of vehicle. The image is taken as an RGB image. RGB stands for red, green, and blue. RGB images are nothing but correspond to the intensity of red, green, and blue lights. The image is first converted into a grayscale image. First, we have converted the RGB (Red, Green, and Blue) image into an HSV (Hue, Saturation, and Value) image, where Hue represents the color type, and the vibrancy is represented by Saturation, which ranges from 0 to 255. The brightness of the color is represented by Value, which ranges from 0 to 255, where 255 represents brightness, and 0 represents darkness. Next, we have split Hue, Saturation, and Value. Then, by taking this Value, we converted the HSV image to a grayscale image. Next, we performed some morphological operations using the kernel. In Fig. 4 and Fig. 5, we have done tophat and blackhat filtering. In a dark background, the object of interest, whose brightness is enhanced by using the tophat filter. The blackhat operation is done by doing the opposite task. In Fig. 6, Gaussian blur has been performed. It is widely used to reduce image noise and reduce unwanted detail. In Fig. 7, thresholding has been done to minimize the background noise by extracting useful information. In Fig. 8, we have joined all the continuous points, which are known as contours, and

respectively, we have to find the possible characters and final contours in Fig. 8 using geometric filtering equations. In Fig. 9-10, we have successfully detected the license plate, and finally, in Fig. 12, we crop the region with the license plate. Cropping the license plate is not done manually. Equations used in this process are:



Fig2. Original image

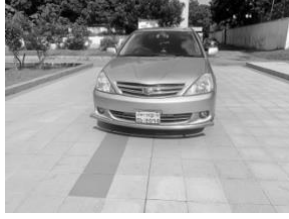


Fig3. Grayscale image



Fig4. Tophat image.



Fig5. Blackhat image.



Fig6. Blur image



Fig7. Threshold image



Fig 7. Contours image



Fig8. Possible character

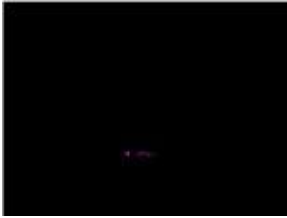


Fig9. Final contours image.

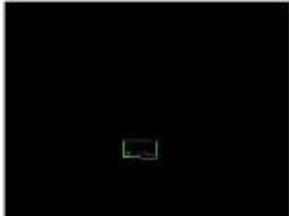


Fig10. Detected image



Fig11. Detected original image.



Fig12. Number plate image

Method 2: CNN a) For language detection

The number plate of Bangladeshi vehicles has a different format compared to the other countries. In our country, we follow 2 two-line format compared to the 1-line format of other countries. In addition, the Bengali characters and their complex lexical structure forced us not to use any public dataset of the license plates from any countries other than Bangladesh. Initially, we must detect the language written on the number plate. This process can be done in the following ways:

Step 1: Train a classifier for language detection

Use a simple classifier like CNN and train on pre-segmented license plate data to distinguish between the visual patterns of Bangla and English characters. It could easily differentiate between those perceiving the complex curves of Bangla characters or the straight lines and angles of English letters.

Step 2: Language Classification:

The trained model would output a label indicating whether the detected license plate text is in Bangla or English. Based on this prediction, we can route the image to the appropriate character recognition process.

a) *Character Recognition*: A model which involves six different layers, including two layers for convolution, two layers for subsampling or pooling, one fully connected layer, and one final classification layer, were used in this experiment. In the pooling layer, the number of feature maps does not change. Sixteen filters are used, which produces 16 feature maps from the second convolutional layer, in the second convolutional layer. Then the second max pooling layer is applied. A fully connected layer with 400 neurons is used, followed by a final classification layer with 18 neurons for 18 different classes. In TABLE I. it is shown that the number of parameters of our model is calculated for 32 x 32 input image.

Table I. Different CNN Layers And Their Parameters

Layer	Layer of operation	Number of feature maps	Feature maps size	Window size	Parameters
C1	Convolution	6	28X28	5X5	156
S1	MaxPooling	6	14X14	2X2	0
C2	Convolution	16	10X10	5X5	2,496
S2	MaxPooling	16	5X5	2X2	0
F1	Fully Connected	400	1X1	N/A	166,400
F2	Fully	18	1X1	N/A	7,218

For the first convolutional layer, the output feature maps size is 28 x 28 for 6 feature maps. The filter mask size is 5 x 5 for both convolutional layers. The number of network parameters used to learn is $(5 \times 5 + 1) \times 6 = 156$, and the one is added with filter dimension because of bias. The number of trainable parameters of the subsampling layer is 0. However, the size of the feature maps was reduced with factor 2 due to a 2x2 pooling mask; therefore, the output of

the first subsampling layer was 14x14 with 6 feature maps. The second convolutional layer's output dimension is $(14-5)+1=10$, 10x10 with 16 feature maps are used and the second subsampling filter mask size is 2 x 2. The learning parameter for second convolutional layer are $((5 \times 5 + 1) \times 6) \times 16 = 2496$. The zero number of network parameters exist in the second subsampling layer. The total number of features per sample from the second sub-sampling layer is $5 \times 5 \times 16 = 400$. Since we have used 400 hidden neurons for this layer, therefore, the number of parameters for the first fully connected layer is $400 \times 16 \times (5 \times 5 + 1) = 166,400$, whereas the amount of the final layer parameter is: $18 \times (400 + 1) = 7,218$. The following code is mentioned below:

```
# Import necessary libraries
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten, Dense

# Define the model
model = Sequential()

# Layer 1: Convolutional Layer (C1) with 6 feature maps and 5x5 filter size
model.add(Conv2D(filters=6, kernel_size=(5, 5), activation='relu', input_shape=(32, 32, 1)))
# Output size: 28x28x6

# Layer 2: Max Pooling Layer (S1) with 2x2 pooling
model.add(MaxPooling2D(pool_size=(2, 2)))
# Output size: 14x14x6

# Layer 3: Convolutional Layer (C2) with 16 feature maps and 5x5 filter size
model.add(Conv2D(filters=16, kernel_size=(5, 5), activation='relu'))
# Output size: 10x10x16

# Layer 4: Max Pooling Layer (S2) with 2x2 pooling
model.add(MaxPooling2D(pool_size=(2, 2)))
# Output size: 5x5x16

# Layer 5: Flatten the output from the convolutional layers
model.add(Flatten())
# Output size: 400 (5x5x16)

# Layer 6: Fully Connected Layer (F1) with 400 neurons
model.add(Dense(units=400, activation='relu'))

# Final Layer (F2): Fully Connected Classification Layer with 18 neurons (for 18 output classes)
model.add(Dense(units=18, activation='softmax'))

# Compile the model
model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])

# Print the model summary to visualize the architecture and parameters
model.summary()
```

The total number of parameters is 1,76,270 for this model.

As English numbers as so many digits so we need to train our model to detect the digit from the license plates. We take the Modified National Institute of Standards and Technology (MNIST) dataset which has a large collection of handwritten digits. The following code is to import TensorFlow and MNIST dataset under the Keras API.

```
import tensorflow as tf
(x_train, y_train), (x_test, y_test) =
    tf.keras.datasets.mnist.load_data()
```

There are four groups train, tests, labels, and images that we separated. x_train and x_test parts contain grayscale RGB codes while y_train and y_test parts contain labels. Matplotlib helps to visualize these numbers.

```
import matplotlib.pyplot as plt
image_index = 8888 #range is up to 60,000
print(y_train[image_index]) #label is 3
plt.imshow(x_train[image_index], cmap='Greys')
```

We get (60000, 28, 28) as a output which convert the image into 28 x 28 pixel size. To work with the Keras API we need to reshape the array into 4-dimensions. So, we use Keras API and implement it in the TensorFlow.

```
#import Keras modules from keras.models import
Sequential from keras.layers import Dense, Conv2D,
Dropout, Flatten, MaxPooling2D
#create model with layers model = Sequential()
model.add(Conv2D(28, kernel_size=(3,3),
input_shape=input_shape))
model.add(MaxPooling2D(pool_size=(2, 2)))
model.add(Flatten()) #fully connected layers
model.add(Dense(128, activation=tf.nn.relu))
model.add(Dropout(0.2))
model.add(Dense(10, activation=tf.nn.softmax))
```

We train our data to fit our CNN model where we will test the data for 10 epoch numbers.

IV. EXPERIMENTAL EVALUATIONS

Fig. 13 shows how our CNN model works in detecting characters.

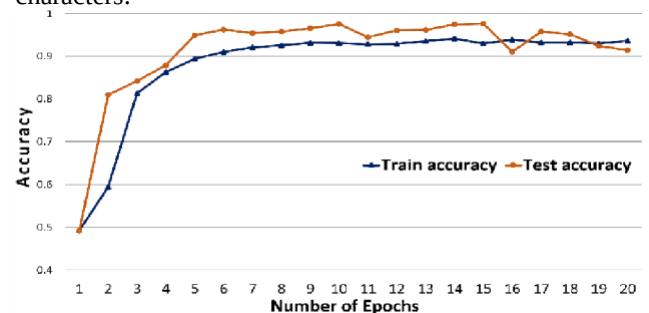


Fig.13: Train vs Testing Accuracy

From this figure, we learn that we have accomplished 93.54% accuracy on the training set and 91.38% accuracy on the test set after 20 epochs. In the very following figure,

we will get to know about the database of the recognized number plate by running our system.

NUMBERPLATE	CURRENTPLATE
1. WB-02-PA-2548	2019-12-02 12:28:01.000044
2. WB-02-PA-2548	2019-12-02 12:28:01.579628
3. WB-02-PA-2548	2019-12-02 12:28:02.200033
4. WB-02-PA-2548	2019-12-02 12:28:03.207033
5. WB-02-PA-2548	2019-12-02 12:28:04.200038
6. WB-02-PA-2548	2019-12-02 12:28:05.000009
7. WB-02-PA-2548	2019-12-02 12:28:06.000009
8. WB-02-PA-2548	2019-12-02 12:28:07.700039
9. WB-02-PA-2548	2019-12-02 12:28:08.000032
10. WB-02-PA-2548	2019-12-02 12:28:09.070032
11. WB-02-PA-2548	2019-12-02 12:28:10.300032
12. WB-02-PA-2548	2019-12-02 12:28:11.000036
13. WB-02-PA-2548	2019-12-02 12:28:12.000036
14. WB-02-PA-2548	2019-12-02 12:28:13.000036
15. WB-02-PA-2548	2019-12-02 12:28:14.200040
16. WB-02-PA-2548	2019-12-02 12:28:15.000042
17. WB-02-PA-2548	2019-12-02 12:28:16.000036
18. WB-02-PA-2548	2019-12-02 12:28:17.000036

Fig. 14. Database of recognized number plates.

V. CONCLUSION

In this paper, we have successfully proposed and implemented a smart parking system leveraging IoT, computer vision, and deep learning techniques to address the growing issue of parking management in urban areas. Our model detects available parking spaces and incorporates vehicle number plate recognition in both English and Bangla using CNN-based YOLOv3 and character recognition models. The system achieved high % accuracy rates of 91% for computer vision and 95% with YOLOv3 in number plate detection, while CNN outperformed traditional OCR methods in character recognition, with a 91.38% accuracy.

Through the development of an Android-based application, drivers are informed of nearby available parking spaces, improving the overall parking experience. Our proposed navigation bot, based on Deep Reinforcement Learning, has demonstrated significant potential by reducing time wastage by 50% compared to traditional methods, and it achieved an impressive 98% accuracy in license plate detection. This improves parking efficiency and paves the way for future advancements.

Looking ahead, we plan to refine the model further by incorporating real-time data collection for a more dynamic system and exploring the use of Blockchain technology to secure user and parking data. This solution presents an opportunity to address the parking crisis in megacities. It can be expanded to include vehicle tracking to assist with stolen vehicle recovery, thus broadening the scope of our smart parking system to contribute to safer and smarter cities globally.

REFERENCES

- [1] A. Khanna and R. Anand, "IoT based Smart Parking System," *2016 International Conference on Internet of Things and Applications (IOTA)*, Maharashtra Institute of Technology, Pune, India, Jan. 22–24, 2016, pp. 1–5, doi: 10.1109/IOTA.2016.7562735
- [2] M. O. Hasan, M. M. Islam, and Y. Alsaawy, "Smart Parking Model based on Internet of Things (IoT) and TensorFlow," *2019 7th International Conference on Smart Computing & Communications (ICSCC)*, Sarawak, Malaysia, 2019, pp. 1–6, doi: 10.1109/ICSCC.2019.8843651

- [3] N. I. Suvon, R. Khan, and M. Ferdous, "Real-Time Bangla Number Plate Recognition using Computer Vision and Convolutional Neural Network," *2024 International Conference on Computer Vision and Machine Learning (ICCVML)*, Dhaka, Bangladesh, 2024, pp. 1–6, doi: 10.1109/IICAIET49801.2020.9257843
- [4] M. Mondal, P. Mondal, N. Saha and P. Chattopadhyay, "Automatic number plate recognition using CNN based self synthesized feature learning," *2017 IEEE Calcutta Conference (CALCON)*, Kolkata, 2017, pp. 378-381, doi: 10.1109/CALCON.2017.8280759.
- [5] N. Abirami and, J. S. L. Jasmine, "Accurate vehicle number plate recognition and real time identification using raspberry pi," *International Research Journal of Engineering and Technology (IRJET)*, vol. 5, pp. 2563–2569, 2018.
- [6] P. Kulkarni, A. Khatri, P. Banga and K. Shah, "Automatic number plate recognition (ANPR) system for Indian conditions," *2009 19th International Conference Radioelektronika*, Bratislava, 2009, pp. 111114, doi: 10.1109/RADIOELEK.2009.5158763.
- [7] N. Vishwanath, S. Somasundaram, M. R. R. Ravi and N. K. Nallaperumal, "Connected component analysis for Indian license plate infra-red and color image character segmentation," *2012 IEEE International Conference on Computational Intelligence and Computing Research*, Coimbatore, 2012, pp. 1-4, doi: 10.1109/ICCIC.2012.6510323.
- [8] 9. P. Chippalkatti, G. Kadam and V. Ichake, "I-SPARK: IoT Based Smart Parking System," *2018 International Conference On Advances in Communication and Computing Technology (ICACCT)*, Sangamner, 2018, pp. 473-477.